SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I 192126068 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	0	0	1	0

Total Marks 33 **/40**

Signature of the Faculty

CO1_1 AT - 1

Name : Ajay Raja

Assesment.

Course : CSA 07 [illegible]

Course Name: Computer Networks

Serial NO: 04

33/40

Questions:

06) Before transmission, the presentation layer compresses a 900 MB file by. 40%. The compressed size calculate the compressed file size, the encrypted file size and the total percentage reduction compared to the orginal file Explain how the Presentation layer

07) An FTP application transfer 3GB of data over a network with an effective throughput of 120 Mbps. The Application layer devides the data into request of 3MB each. calculate the total number of request generated and estimate the total transmission time

08) A packet experience the following processing delay at different OSI layers: Application layer 4 ms, Transport layer = 8ms, Link layer = 2ms, physical layer 1ms. Calculate the total end to end delay experience by the packet and explain how.

b) Presentation Layer - Compression and Encryption

Given:

- Orginal File size = 900MB
- compression = 40 %
- Encryption overhead = 5% of compressed size

Step 1: calculate the compressed file size

$$\text{compressed size} = 900 \times \frac{100 - 40}{100}$$

$$= 900 \times 0.60$$

$$= 540\text{MB}$$

Step 2: Calculate the encrypted file size, Encryption add 5% overhead to the compressed file size

$$\text{overhead} = 540 \times \frac{5}{100} = 27\text{ MB}$$

$$\text{encrypted size} = 540 + 27$$

567 MB

Step 3: Calculate the total percentage reduction:

$$\text{Reduction} = 900 - 567 = 333\text{ MB}$$

$$\text{Percentage Reduction} = \frac{333}{900} \times 100 \Rightarrow 37\%$$

Role of the presentation layer:

① compression : Reduce file size, saving bandwidth and transmission time

② Encryption : Protects data from unauthorized accen during transmission, like converts the plain text to cypher text.

③ Data Formatting : Converts data into a suitable Format for communication between different systems.

④ Decompression and Decryption : The receiving system can restore the orginal data and read it.

Thus, the prosentation layer improoves transmision efficiency through comporession and enhances securit through encryption.

Because, when we didn't encrypt the file we have that leads to a mal practice or hacker can logged in and destroy it forever.

Application Layer - FTP Data transfer :

* FTP is a file transfer protocal used to transfer a file from a computer.

Given :

- Total data = 3GB
- Effective throughput = 120 Mbps
- Request size = 3MB

Assume 1GB = 1000 MB and 1byte = 8 bits

Step 1 : Calculate the total number of request.

Total Data = $3 \times 1000 = 3000$ MB

Number of Request = $\frac{3000}{3}$

1000 request

Step 2 : Calculate the total transmission time

Convert the total data into megabits

$3000 \times 8 = 24000$ megabits

Transmission time:

$$\text{Time} = \frac{\text{Total data in Megabits}}{\text{Through put}}$$

$$= 24000 / 120 = 200 \text{ seconds}$$

Therefore, the estimated transmission time is 200 seconds (3minutes 20 seconds), Assuming constant throughput and excluding protocal overhead and Additional delays.

Role of the Application layer:

User Services:

Provides network services through application such as FTP, email and web browser.

File transfer:

FTP enables users to upload and download files over a network

Communication:

Supports Communication between application on different devices.

Authentication:

FTP can use username and passwords to control access, although traditional FTP does not encrypt credentials.

Error handling:

FTP reports transfer errors and supports responses that help application identify failures and retry operations.

Conclusion: The Application Layer provides services directly to user Application. FTP facilitates file transfer, while it protocol response and the underlying transport layer help detect errors and support reliable delievery.

Calculation of End to End Delay in the OSI Model:

Given:

- Application layer delay = 4 ms
- Transport layer delay = 3 ms

- Network layer delay = 5ms
- Data link layer delay = 2ms
- Physical layer Delay = 2ms
- Propagation delay = 18ms
- Transmision delay = 12ms

Step 1: Calculate total processing delay:

$$D_p = 4 + 3 + 5 + 2 + 1$$

$$= 15ms$$

Step 2: Calculate the total end-to-end delay

$$D_{total} = D_p + D_{prop} + D_{trans}$$

$$= 15 + 18 + 12$$

$$= 45ms.$$

Contribution of each OSI layer:

1. Application layer: Provides network services to application such as web browser and email
2. Transport layer: Provides end to end communication, segmentation, and error recovery when supported by the protocol

Step 2 : Calculate the total number of frames.

Convert useful data into bytes.

80 MB = 80,000,000 bytes

$$\text{Frames} = \frac{80,000,000}{1520}$$

$$= 52631.58$$

Since fraction of a frame is not possible, round up;

52632 Frames

Step 3: calculate the total overhead transmitted

Total overhead = 52632 × 80

= 4,210,560 bytes

Therefore, the total overhead is Approximately equal to 4.21 MB

Step 4: calculate transmission efficiency:

Transmission Efficiency is the percentage of transmitted data that carries useful information

3. Network layer: Determine the route and forms packets toward the direction

4. Data link layer: Organize data into frames and handles local delivery

5. Physical layer: Transmit bits to cables, fibre or wireless signals.

⑨ Calculation of frames, protocal overhead and transmision Efficiency:

Given:

- useful data: 80MB
- Frame size: 1600 bytes.
- Protocal overhead per frame = 80bytes

Assume = 1MB = 10^6 bytes

Step 1: Calculate useful data per frame:

useful data per frame = 1600 - 80

= 1520 bytes

$$\text{efficiency} = \frac{\text{useful data}}{\text{Total transmitted data}} \times 100$$

Total transmitted data = 52682 × 1600

= 84,211,200 bytes

$$\text{Efficiency} = \frac{80,000,000}{84,211,200} \times 100$$

Efficiency = 95%

Effect of protocal overhead on network performance

1. Increase transmitted data: Headers and trailers add extra bytes to each frame
2. consume bandwidth: overhead use network capacity that could otherwise carry useful data
3. may increase delay: more data must be transmitted, potentially increasing transmission time
4. Supports reliable communication: Protocal overhead carries addressing, sequencing and error detection..